Mosaic

A photo album portfolio

CSE 4500 — Platform Computing

Solomon Smith · Isaac Munoz · Dominic Arrezola · Charles Phan

Live demo: *(filled in after Render deploy)*

The problem

The course covered HTML, CSS, JavaScript, Express.js and database management as separate modules.

> The final project asks us to combine all of them into one application.

We chose a **photo album portfolio** because it exercises every layer:

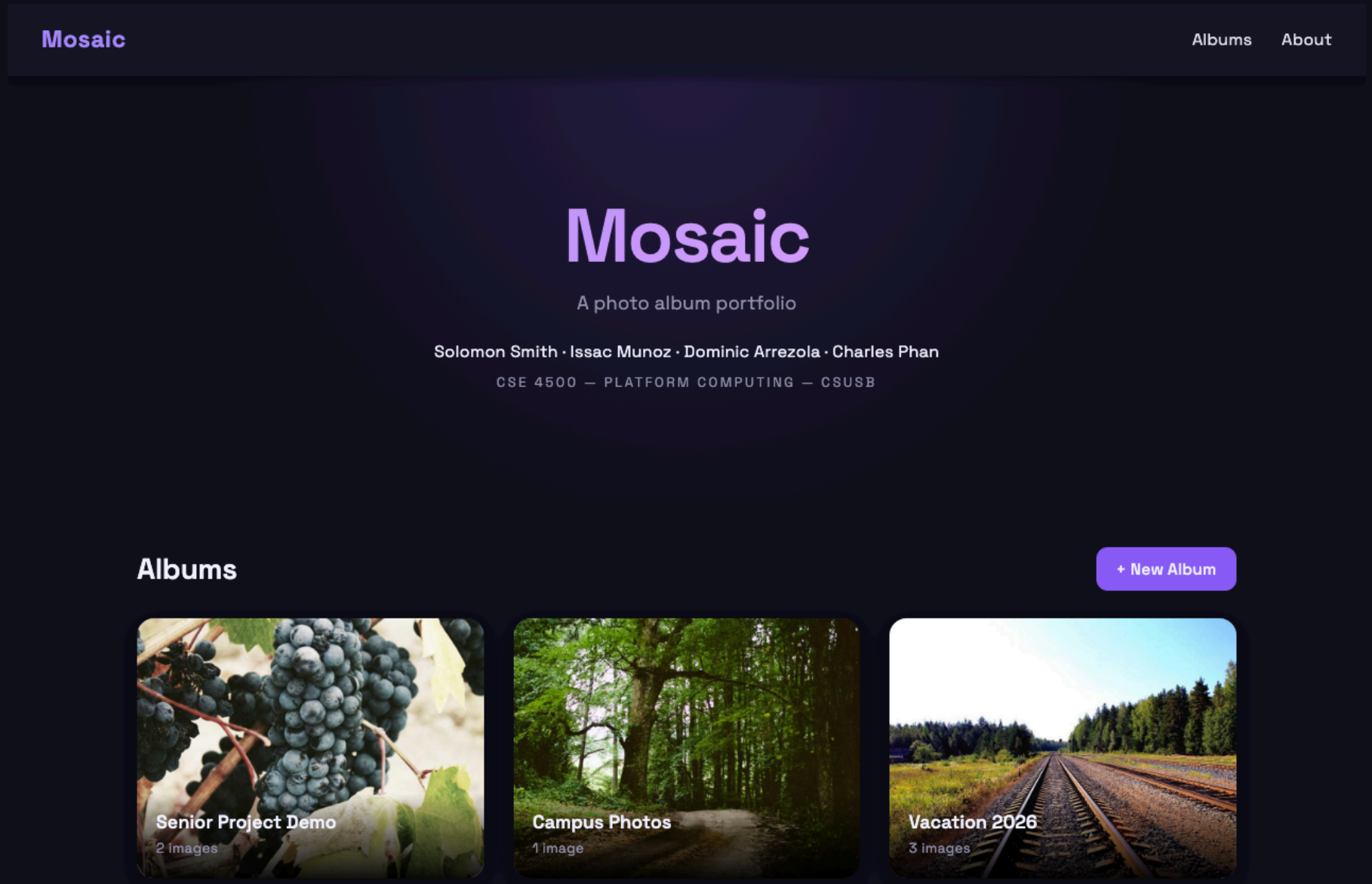
- Styled, responsive frontend
- REST routes
- Multipart file uploads
- Relational schema with a foreign-key cascade
- Persistent file storage on deploy

Architecture

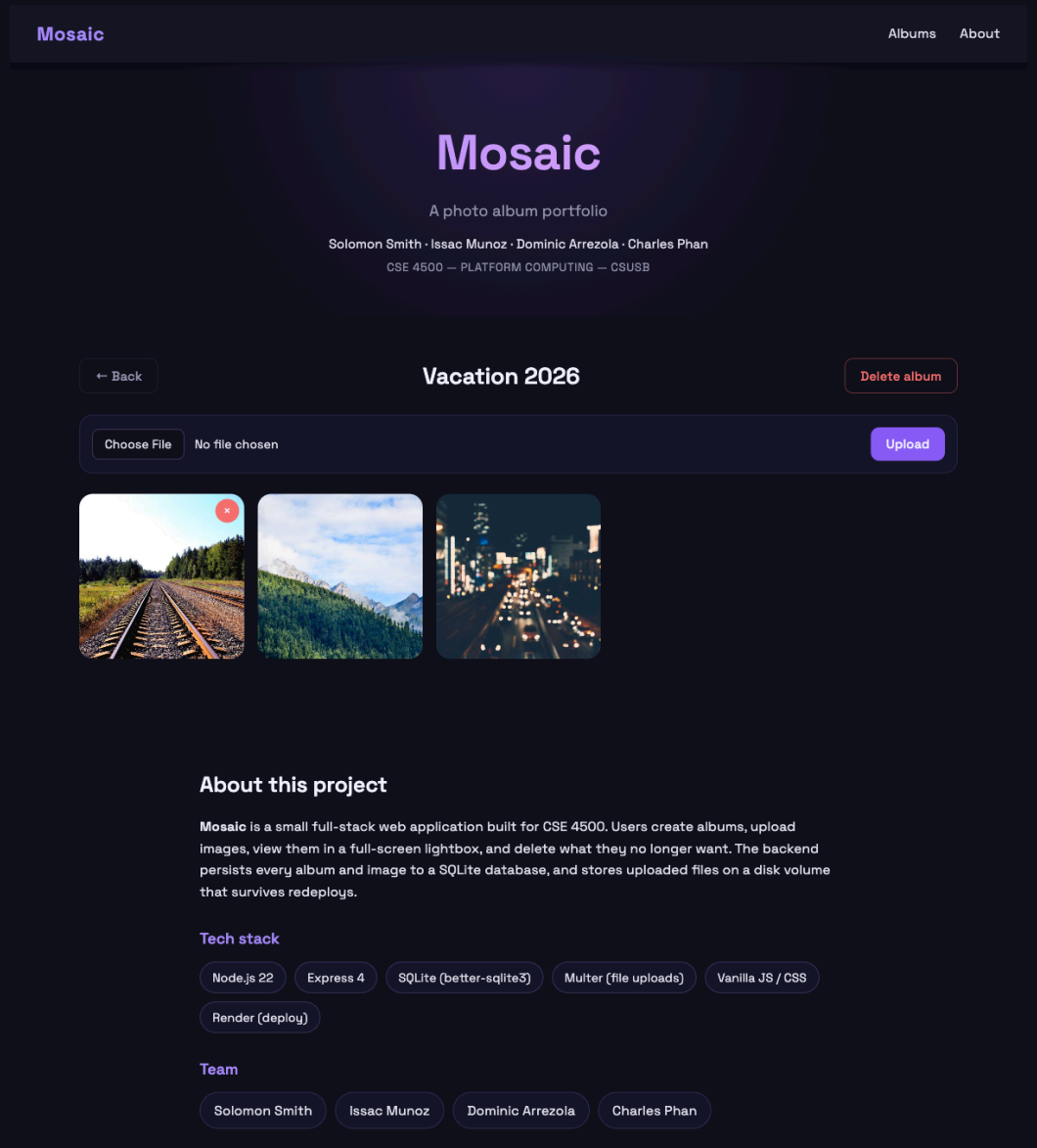
```
Browser  ──HTTP──>  Express (Node 22)
                    │
                    ├─ GET    /api/albums          → SQLite
                    ├─ POST   /api/albums          → SQLite
                    ├─ DELETE /api/albums/:id      → SQLite + fs.unlink
                    ├─ POST   /api/upload          → multer + SQLite
                    ├─ DELETE /api/images/:id      → SQLite + fs.unlink
                    ├─ /uploads/*                  → static images
                    └─ /                           → static SPA
```

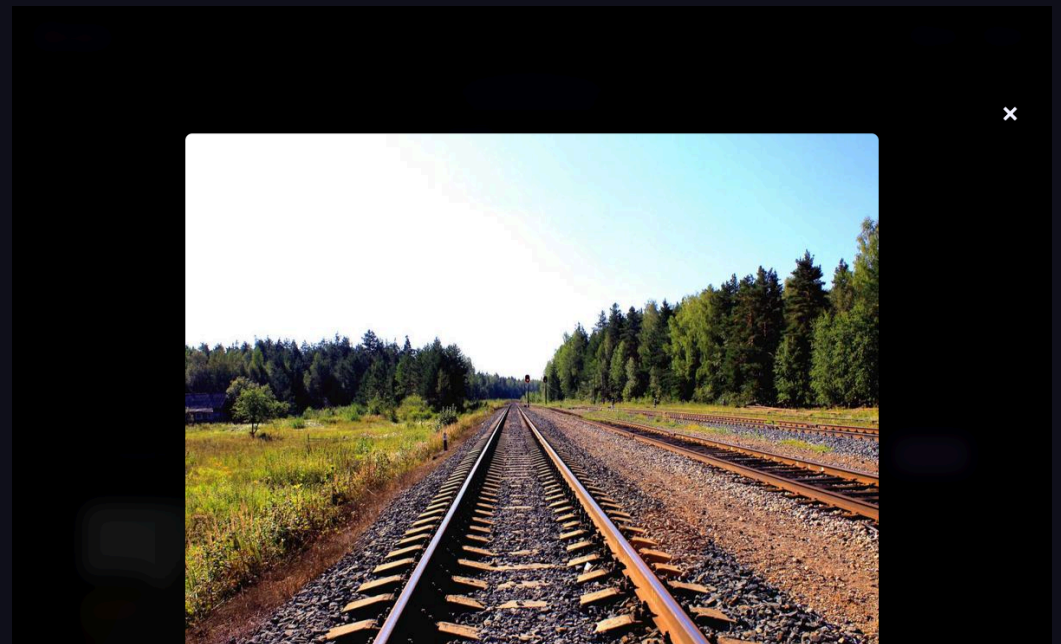
Two tables: `albums` and `images`, joined by `images.album_id` with `ON DELETE CASCADE`.

Live demo


Mosaic
Albums
About
Mosaic
A photo album portfolio
Solomon Smith · Issac Munoz · Dominic Arrezola · Charles Phan
CSE 4500 — PLATFORM COMPUTING — CSUSB
Albums
+ New Album
Senior Project Demo
2 images
Campus Photos
1 image
Vacation 2026
3 images

Live demo — gallery + lightbox





Tech stack

Layer	Choice
Runtime	Node.js 22 (pinned in `.node-version`)
Server	Express 4
Database	SQLite via `better-sqlite3` (synchronous)
Uploads	Multer (10 MB cap, image-only filter)
Frontend	Vanilla JS + CSS (no framework, no innerHTML)
Deploy	Render Blueprint with 1 GB persistent disk

Validation

The backend returns JSON error envelopes the frontend shows as toasts:

Scenario	Status
Duplicate album name	409
Empty album name	400
Non-image upload	400
Upload > 10 MB	400
Upload to nonexistent album	404
Delete missing image / album	404

If the album check fails *after* multer has already written a temp file, the route unlinks the orphan before responding.

Lessons learned

- **Native modules pin the runtime.** `better-sqlite3` is compiled against a specific Node ABI — `.node-version` documents the pin.
- **SQLite foreign keys are off by default.** Without `PRAGMA foreign_keys = ON;` cascading deletes silently no-op.
- **Multer writes before your route runs.** Validate cleanup paths.
- **`textContent` beats sanitizing `innerHTML`.** Structural XSS prevention with zero dependencies.

Team

Mosaic is a joint effort by all four team members.
Every member contributed across the stack.

- **Solomon Smith**
- **Isaac Munoz**
- **Dominic Arrezola**
- **Charles Phan**

Questions?

Repository: github.com/.../cse-4500-final-project

Live demo: *(Render URL)*

Thank you.